ZCAS UNIVERSITY

SCHOOL OF COMPUTING AND APPLIED SCIENCES

BACHELOR OF INFORMATION TECHNOLOGY/BACHELOR OF NETWORK ENGINEERING

CIT1142 EMERGING TRENDS IN ICT

END OF SEMESTER FINAL EXAMINATION

Monday 8th June 2026

09:00 – 12:00 HOURS

TIME ALLOWED: **WRITING – THREE (3) HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Section B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer the following question.

QUESTION 1.

Read the following then answer the questions that follow.

John Smith, an investigator specializing in computer forensics, has been assigned to a case involving a suspected data breach at ZSIC Corporation. The company's confidential information, including customer data and financial records, may have been compromised. John Smith begins the investigation by examining the company's computer systems.

a) What steps should John Smith take in the initial assessment phase of the computer forensics investigation to ensure the preservation of digital evidence? **(8 Marks)**

b) As John Smith identifies potential evidence related to the data breach, what tools and techniques can be employed to properly collect and preserve digital evidence without altering its integrity? **(8 Marks)**

c) Explain why maintaining a strict chain of custody is crucial in computer forensics, and what measures John Smith should implement to ensure the integrity of the digital evidence throughout the investigation process. **(8 Marks)**

d) If some of the compromised data appears to be deleted or encrypted, what methods and tools can John Smith use to recover and analyse the timeline of events leading up to the data breach? **(8 Marks)**

e) During the investigation, John Smith discovers evidence that may be admissible in court. What legal considerations and best practices should he follow to ensure that the collected digital evidence is admissible and can withstand legal scrutiny? **(8 Marks)**

[Total: 40 Marks]

SECTION B

Answer ONLY three questions from this section.

QUESTION 2.

In the rapidly evolving landscape of global finance, banks are facing unprecedented challenges in safeguarding their customers' assets against a rising tide of fraudulent activities. Bank fraud has become a pressing issue, with perpetrators employing increasingly sophisticated methods to exploit vulnerabilities in financial systems. As a junior analyst at a cybersecurity firm specializing in financial fraud detection, you are tasked with examining trends in bank fraud across cities worldwide to enhance preventative measures and protect customers.

a) What geographical trends do you observe in the prevalence of bank fraud across major cities globally, and are there specific regions showing higher susceptibility to these activities? **(5 Marks)**

b) How have bank fraud techniques evolved in recent years, and what emerging methods are being employed by perpetrators to exploit vulnerabilities in different cities?

(5 Marks)

c) Are there discernible patterns in terms of demographics being targeted by bank fraudsters in various cities, and how can this information be utilized to tailor preventive measures?

(5 Marks)

d) How can international collaboration and information sharing between financial institutions and law enforcement agencies be improved to mitigate the cross-border nature of modern bank fraud?

(5 Marks)

[Total: 20 Marks]

QUESTION 3.

A large e-commerce company, Amazon, is looking to enhance its business intelligence and customer insights through data mining. The company has collected a vast amount of data over the years, including customer purchase history, website interactions, and feedback. The management team is eager to leverage data mining techniques to uncover valuable patterns, trends, and relationships within this data to make informed business decisions and improve the overall customer experience.

a) Explain how you would approach the data preparation phase for Amazon's e-commerce data before applying data mining techniques. **(5 Marks)**

b) List and discuss potential challenges that might arise in cleaning and transforming the diverse data sets collected by Amazon. **(5 Marks)**

c) Which data mining techniques do you think would be most suitable for analyzing customer purchase history and predicting future buying behavior for Amazon? **(5 Marks)**

d) How can clustering algorithms be applied to segment Amazon's customer base for targeted marketing strategies?

(5 Marks)

[Total: 20 Marks]

QUESTION 4.

You are a computer science specialist tasked with researching and understanding various aspects of computer applications in web services. Your IT Manager has assigned you a project to explore the fundamentals, technologies, and trends related to web services in the context of computer applications. To complete this project, you will need to conduct interviews with other international experts in the field, conduct research, and analyze the current state of web services in computer applications.

a) Explain to the IT Manager how the use of APIs (Application Programming Interfaces) relates to web services, and what role they play in modern web applications. **(5 Marks)**

b) Discuss the concept of REST (Representational State Transfer) and explain its significance in web service development for computer applications. **(5 Marks)**

c) How do SOAP (Simple Object Access Protocol) and RESTFUL web services differ in terms of communication and data exchange in computer applications? **(5 Marks)**

d) Explain the significance of microservices architecture in the context of web services, and how it relates to computer applications.

(5 Marks)

[Total: 20 Marks]

QUESTION 5.

At a mid-sized company, the IT department has implemented an Intrusion Detection System (IDS) to enhance the organization's cybersecurity posture. The IDS is responsible for monitoring network traffic and identifying potential security incidents. One day, the IDS triggers an alert indicating a possible intrusion attempt. The IT team assigns Monde, a junior cybersecurity analyst, to investigate the alert and determine whether it's a genuine security threat.

a) Explain the steps Monde should take to investigate the IDS alert thoroughly. Outline the key procedures and tools she might use to determine the nature and severity of the potential intrusion. **(5 Marks)**

b) Define the concepts of false positives and false negatives in the context of intrusion detection systems. How can Monde distinguish between a false positive and a genuine security incident during her investigation? **(5 Marks)**

c) If Monde confirms a security incident based on the IDS alert, what should be her immediate response actions? Outline the steps she should take to contain and mitigate the potential threat to the organization's network. **(5 Marks)**

d) Discuss the concept of IDS tuning and optimization and provide examples of adjustments Monde might make to enhance the system's ability to detect and respond to potential intrusions. **(5 Marks)**

[Total: 20 Marks]

END OF SEMESTER FINAL EXAMINATION